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Expedite Multi-Scenario Assisted History Matching with AI-Enabled Reservoir Surrogate Modeling

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Abstract

Production from multi-bench unconventional development is time-consuming to model in physics-based simulators, requiring multiple runs. Each simulation takes minutes to hours, even with enhanced cloud computing power. This limits history matching to a single deterministic solution, as multi-scenario algorithms need significantly more simulations. An AI-based surrogate model expedites forecasting, enabling efficient multi-scenario algorithms.

The AI surrogate relies on a large dataset of high-fidelity physics-based simulations created using modern high-performance computing (HPC) infrastructure. Simulations were based on Latin-hypercube sampling of over 100 parameters, covering geology, reservoir properties, well placements, and completion designs. Results were processed on HPC to extract features and train a surrogate model using advanced AI techniques. The model was iteratively trained to meet user-defined error metrics. The final surrogate was integrated with an automated history matching (AHM) package using a Gauss-Newton trust region optimizer to obtain multiple scenarios that align with field production.

Developing a surrogate model with the desired fidelity was a challenging task. Reservoir models include several highly non-linear input-output relationships from various physical processes. Training a surrogate with an acceptable level of fidelity requires both a massive dataset and specialized AI techniques to enforce physical relationships in deep learning algorithms. The latter was critical to obtain correct physical trends from the surrogate model predictions. The final surrogate model's production forecasts accurately match the trends and magnitudes of the computationally expensive, high-fidelity numerical simulations across several real examples of multi-well and multi-bench developments. The surrogate model integration within the AHM package was straightforward and enabled the efficient determination of multiple scenarios in about one hour for a typical unconventional case. This expedites the determination of plausible scenarios, empowering end-users to better optimize the development plan.

A fast reservoir surrogate model significantly reduces the cycle time for using physics-based models and workflows, capturing subsurface uncertainty more holistically. The surrogate model workflow benefits

from standard features of machine learning systems, including interpretability and confidence scores, providing the user with richer information before making decisions. In particular, the work demonstrates that determining multiple reservoir scenarios matching production history is practically achievable by leveraging cutting-edge AI technologies.

Introduction

Physics-based production modeling for multi-well, multi-bench unconventional reservoirs has evolved significantly as more field data have become available. It is well established that single-well models are insufficient to capture the complex well interactions commonly observed in unconventional plays. This necessitates the development of computationally intensive models incorporating multiple wells to account for interactions in both vertical and horizontal directions. Consequently, executing such studies can require several weeks to months, even with high-performance computing resources.

Recent advances in subsurface diagnostics have yielded valuable insights into the hydraulic dimensions of fractures during stimulation, along with some qualitative understanding during production ([Liang et al. 2022](#); [Wilkinson et al. 2023](#); [Manchanda et al. 2023](#); [Li et al. 2023](#); [Han et al. 2023](#)). However, significant uncertainties remain in key inputs such as fracture conductivity distribution, long-term fracture dynamics, and multiphase relative permeability behavior. These uncertainties necessitate model calibration through history matching to achieve predictive capability. Moreover, interpreting these measurements is often challenging, non-unique, and largely qualitative due to the presence of multiple wells and the complex flow dynamics characteristic of unconventional reservoirs.

To accelerate modeling workflows, surrogate modeling and machine learning (ML) techniques have been applied to both hydraulic fracture growth and production modeling. ([Gupta et al. 2023](#)) proposed a framework using Recurrent Neural Networks (RNNs) trained on 1,000 single-well hydraulic fracture simulations, though its application was limited to a small geographic area. ([Wang et al. 2022](#)) developed a deep learning model based on U-net architecture trained on 2D images of completed fracture simulations. [Pradhan et al. \(2022\)](#) introduced a physics-informed production forecasting model for the oil phase, leveraging deep learning trained on pre-run reservoir simulations. ([Gala et al. 2023](#)) extended this concept basin wide, eliminating the need for additional simulations or AI training during deployment by leveraging recent advances in AI/ML. The AI/ML algorithms developed in their work capture the highly non-linear relationships between hundreds of model inputs and the full time-series production outputs of each well.

This paper summarizes the methodology and integrates it with a multi-scenario assisted history matching process to rapidly generate multiple history-matched scenarios. By accelerating production forecasting by orders of magnitude, the surrogate model enables the use of assisted history matching algorithms ([Forouzanfar et al. 2021](#)) that are typically computationally prohibitive at full scale, as the search for multiple solutions significantly increases the number of required forecasts. The approach is demonstrated on a synthetic multi-bench unconventional development later in this paper.

Dataset Creation and Post-Processing

The physics-based flow simulations used to generate the dataset were conducted using a fully parallel, in-house reservoir simulator deployed on a high-performance computing cluster. In this study, geological features, fluid properties, completions and fracture characteristics, well spacing, and flowing conditions were systematically varied to construct a comprehensive dataset representative of multi-well, multi-bench unconventional developments. This process resulted in over 150 input parameters, which were sampled using the Latin Hypercube method. A high-level overview of these input parameters is provided in [Figure 1](#). It is important to note that, in all simulation cases, wells begin production within a few days of one another. Therefore, future work is needed to incorporate the effects of prior depletion and parent-child interference. Significant effort was made to ensure that the input parameter ranges reflect field-representative conditions.

Geology properties for every bench • Reservoir porosity • Reservoir horizontal and vertical permeability • Reservoir fluid saturations • Rock properties (compaction, compressibility etc.) • Pore pressure • Reservoir temperature • Reservoir thickness	Fracture properties for every bench • Fracture geometry parameters • Fracture conductivity distribution parameters • Fracture compaction and compressibility parameters • Water saturation in and around the fractures • Fracture orientation • Fracture locations along the well
Fluid properties for every bench • Multi-phase relative permeability parameters • PVT properties parameters • Saturation pressure	Operational properties for every bench • Number of wells • 3D location of individual well • Stage length • Number of fractures per stage • Individual well flowing bottomhole pressure • Well orientation

Figure 1—Categorized Input Parameters Used to Generate the Training Dataset

AI/ML training to create a robust surrogate model

The dataset described in the previous section was preprocessed, and scenarios were filtered to construct a balanced dataset representative of realistic production profiles. A machine learning (ML)-based surrogate model was then trained on this dataset for application across a wide range of unconventional development scenarios. To build a robust surrogate, Science-Based AI (SBAI) techniques were employed. These included physics-informed loss functions and various state representation learning approaches designed to capture multiphase flow behavior, well interference, and spatially varying fracture properties. The methodology used to develop the SBAI model was detailed in a previous study (Gala et al. 2023) and is summarized in Figure 2, which highlights the significant accuracy gains achieved using SBAI.



Figure 2—Actual versus prediction plot for (a) the initial version of the surrogate model, which did not incorporate any science-based AI techniques (left) and (b) the final version of the surrogate model, which incorporated science-based machine learning techniques (right) Model (y-axis) is the result of the surrogate model, and Actual (x-axis) is the underlying simulation result. Full time-series data is plotted

Surrogate Model Validation

Extensive model validation was presented in a previous study (Gala et al. 2023) and conducted across three levels. First, numerical accuracy was evaluated using standard error metrics on holdout test sets to ensure fidelity across full time-series predictions. Second, physical consistency was verified through parameter sensitivity analyses and explainability tools (e.g., SHAP), which confirmed that the model reproduced known physical trends—such as the impact of porosity or bubble point on oil and gas production—even for inputs not explicitly included in the training set. Third, application-based validation was performed by integrating the trained surrogate into workflows such as history matching, which is discussed later in this paper. These use cases demonstrated that the model's recommendations aligned with real-world engineering

intuition, particularly in capturing interference effects across wells and benches. Figure 3 further illustrates that Science-Based AI captures physical trends (e.g., well interference), whereas a traditional neural network of the same size falls short in reproducing this behavior. For more details on physics integration, model training, and benchmarking experiments, refer to (Gala et al. 2023).

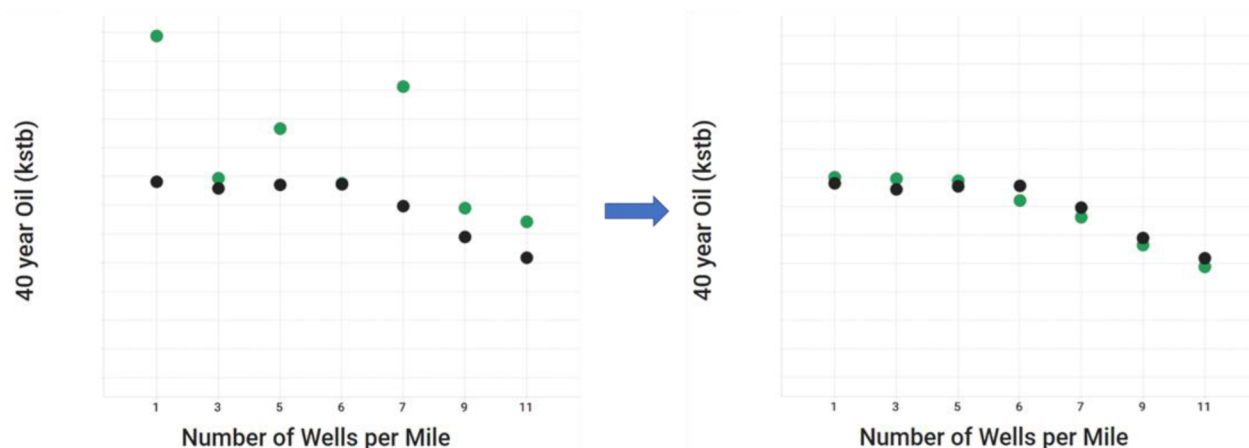


Figure 3—Average production per well at the end of 40 years plotted vs number of wells per mile. Synthetic simulation data is shown in black. Predictions from AI models are shown in green. Left plot is using traditional AI techniques, Right plot is using Science based AI techniques developed as part of this study.

Gauss-Newton history matching algorithm

The history match process is achieved by a Gauss-Newton optimizer with an adaptive trust region method (Al-Rukabi and Forouzanfar, 2019). The outcome from the optimization process is further processed by Gaussian mixture clustering for characterizing the solution space with probabilities.

Gauss-Newton trust region optimization

A reliable surrogate model enables history matching to be performed efficiently. History matching is a process of reducing the difference between the forward model prediction and field production data iteratively through manual adjustments of model parameters or an optimization algorithm. In this case, we integrate the surrogate model and a proprietary assisted history match (AHM) package to arrive at potential solutions automatically.

Our history match configuration can easily involve over 100 parameters (rock properties, fracture geometries, fluid properties, etc. over multiple flow units). Given the high dimensional nature, an ensemble-based multi-scenario approach is appropriate. For this workflow, we use a Gauss-Newton optimizer for finding potential solutions (Al-Rukabi and Forouzanfar 2019).

At every iteration of the Gauss-Newton optimizer, a quadratic approximation of the objective function is formed for each solution trial by an ensemble of perturbations. Each quadratic approximation is then optimized within the adapted trust region as a sub-problem and the solutions to the sub-problems are then carried forward to the next iteration. This process for each solution trial is repeated to the end of iterations as defined or until convergence (Al-Rukabi and Forouzanfar 2019).

At the end of the Gauss-Newton optimization process, we arrive at the final version of the main solution trials in addition to the generated perturbation trials. Together, we obtain an ensemble of trials that can define the solution space. Although the main trials can be perceived as potential multi-scenario solutions, it is still challenging to evaluate each one of them and make decisions without clearer understanding of uncertainty.

Gaussian mixture clustering

The multi-scenario solution space can be further characterized with uncertainty quantification through Gaussian mixture clustering, a clustering technique that leverages Gaussian mixture modeling.

Gaussian mixture modeling assumes the population density is of a mixture of multiple Gaussian distributions. A Gaussian mixture model is parameterized by the mean and covariance of each Gaussian, and weight associated with each Gaussian. These parameters are obtained by the expectation maximization algorithm. The algorithm works as follows:

1. Initialize the density function parameters (means, covariance, and weights)
2. Calculate the probability of each data point belonging to each cluster and collect the total probability (responsibility) for each cluster
3. Update the density function parameters for each Gaussian using total collected responsibilities
4. Repeat the updating process until the parameters stop changing significantly

To apply Gaussian mixture clustering on the ensemble of trials, we follow the steps described as follow:

1. Compute the global errors for all trials (the final version of the main solution trials plus the perturbation trials)
2. Eliminate trials with large global errors
3. Perform standardization and principal component analysis
4. Perform Gaussian mixture clustering on principle components on a range of number of clusters
5. Select the optimal number of clusters at the highest Silhouette score (a minimum of 0.5 is required to define clusters)

If meaningful clusters are formed, we then proceed to evaluate the means of clusters as potential distinct solutions rather than the raw outcome from AHM. The resulting number of clusters is usually significantly smaller than the number of trials initialized for the history match process, indicating the solution space is well-defined given all the trials. Additionally, the weight of each cluster can be perceived as the probability of the corresponding solution being the true solution under the current history match parameter search space.

In addition to providing a probability density function for the solution space, Gaussian mixture clustering also helps the HM process reach solutions faster as a side benefit. In theory, all trials will converge to local minima with sufficient iterations. However, the number of iterations required for all trials to converge is difficult to determine up-front in practice and might be computationally expensive. Applying Gaussian mixture clustering provides an indication of what the solutions are even if the trials have not converged yet. [Figure 4](#) is a conceptual illustration of how Gaussian mixture clustering helps define the solution space efficiently with a simplified example (perturbation trials are not visualized). In this one-dimensional example, the objective function has three local minima. Twenty trials are initialized uniformly ([Figure 4a](#)). At an early stage of optimization ([Figure 4b](#)), the trials are far from convergence and clusters cannot be formed. As they progress further, clusters are formed, and the means of the Gaussian clusters are very close to minima ([Figure 4c](#)). The dots with dashed outlines represent means of clusters (We eliminate trials with large global errors before clustering). Eventually, all trials converge to local minima with sufficient iterations ([Figure 4d](#)). If clustering is formed at this stage, the means would be very similar to the means from the previous stage.

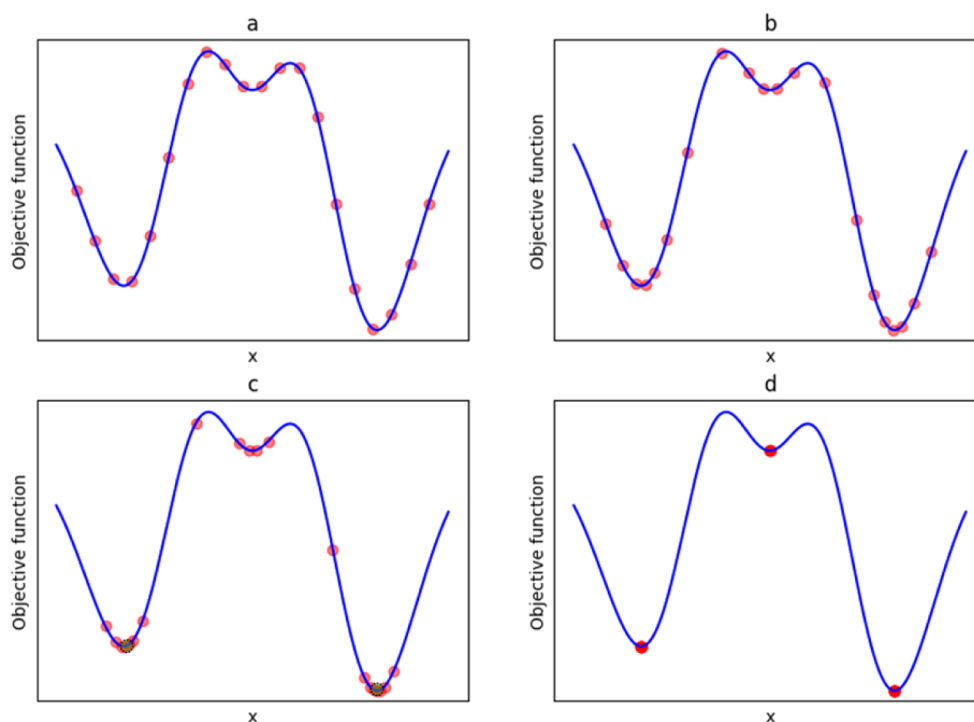


Figure 4—Conceptual illustration of Gaussian mixture clustering applied to a one-dimensional optimization problem: a) initialization – trials are initialized uniformly; b) early stage of optimization – clusters are not formed; c) intermediate stage of optimization – clusters can be formed from trials with lower errors; cluster means are close to true solutions; d) all trials converge to local minima.

Application to multi-bench history matching

The history matching algorithm described above requires numerous forward model evaluations, making it computationally impractical to explore multiple solutions when coupled with a full-physics reservoir simulator. In contrast, the surrogate model presented in this study can generate production forecasts within seconds, significantly increasing the number of scenarios that can be evaluated during the history matching process.

This capability is demonstrated using a synthetic two-bench development example. For confidentiality reasons, the reservoir properties and well configurations do not reflect those of an economically viable development. A gun barrel view of the model is shown in [Figure 5](#), featuring five wells in Bench B2 and eight wells in Bench B3. These wells were completed with overlapping elliptical fractures to ensure interaction between wells. Fluid, reservoir, and geological properties were randomly selected within the bounds of the surrogate model's training data. A full-physics reservoir simulation model was constructed to serve as the reference (ground truth), providing one year of historical production data. The surrogate model was then run using the same input parameters to assess its fidelity relative to the simulator.

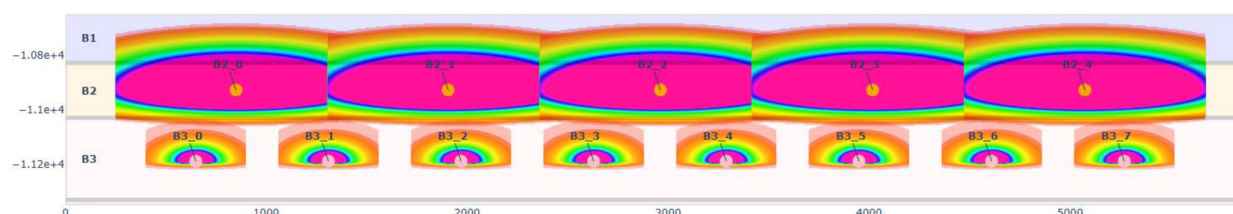


Figure 5—Gun barrel view of the synthetic case studied. The case has 5 wells in bench B2 with fractures that overlap and 8 wells in bench B3 whose fractures grow to touch the bottom of the fractures of bench B2. This leads to fracture interactions and dynamics that need to be appropriately captured to accurately forecast the production of the different wells.

Figure 6 presents 40-year production forecasts from both the simulator and the surrogate model for two representative wells: an edge well (B2_0) and an interior well (B3_1). Due to the symmetric nature of the problem and for brevity, only these two wells are discussed, as their behavior is representative of the broader well set. The largest discrepancy is observed in the cumulative oil production of well B3_1, which is attributed to its extremely high water cut (greater than 0.99), making it challenging for the surrogate to accurately predict oil production. A single surrogate model was trained to predict cumulative oil, water, and gas production. The loss function assigned approximately equal weights to oil and water, so when one phase dominates, the other becomes underrepresented during training, leading to larger forecast errors. Similarly, the gas production error for B3_1 is due to its low gas output—approximately ten times lower than that of B2_0 for similar liquid production. Nevertheless, even under these challenging conditions, the surrogate model correctly predicts low oil production, demonstrating its robustness even in scenarios that deviate from typical development conditions.

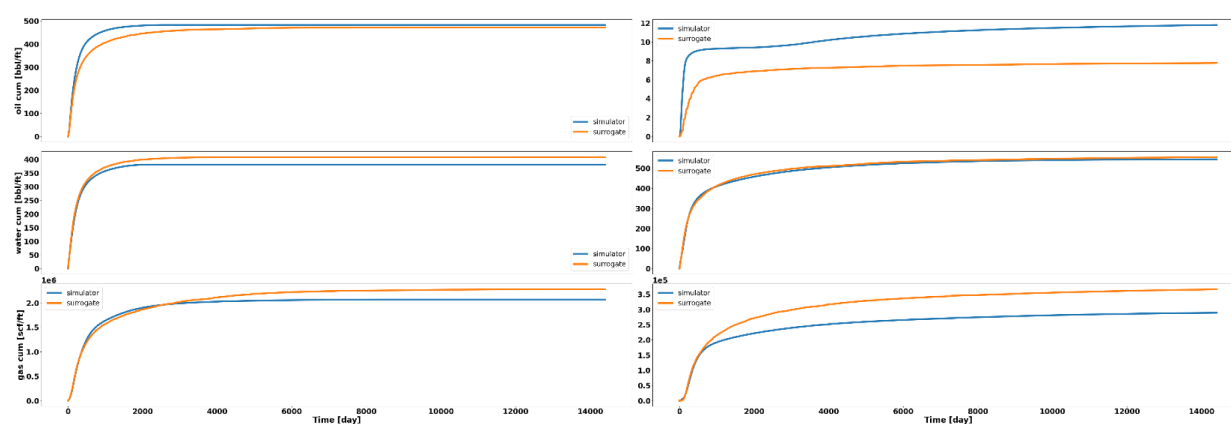


Figure 6—Comparison of cumulative oil, water, and gas production between the surrogate model and the reservoir simulator, which serves as the reference solution for history matching. For well B2_0 (left), the surrogate predicts all three production phases with approximately 10% Mean Absolute Percentage Error (MAPE). In contrast, for well B3_0 (right), only the water phase is forecasted within 10% MAPE. The oil and gas production from B3_0 are at least an order of magnitude lower than those from the B2 wells, despite similar water production levels. This disparity presents a challenge for the surrogate model's accuracy in low-production regimes. However, the impact on the overall forecast remains limited, as total oil and gas production is primarily driven by the B2 wells.

Table 1 reports the phase-specific Mean Absolute Percentage Error (MAPE), averaged over the one-year historical period. For both wells, the MAPE is approximately 10%, except for oil and gas in well B3_1, as discussed above.

Table 1—Mean Absolute Percentage Error (MAPE) for each production phase in representative wells B2_0 and B3_1. The errors for phases that significantly contribute to total production are approximately 10%.

Well	Oil [%]	Water [%]	Gas [%]
B2_0	13	6	5
B3_1	54	4	23

The Gauss-Newton algorithm described earlier was used to identify 30 distinct history match solutions, each supported by 20 ensemble members. One edge and one interior well per bench were matched across all three production phases, with reduced weighting on oil production in Bench B3 due to anticipated measurement uncertainty. In practice, separating oil and water in such high water cut environments is difficult, reducing confidence in oil production data. The algorithm was run for 20 main iterations to ensure convergence. Fracture and relative permeability parameters in Benches B2 and B3 were treated as uncertain

and varied during the assisted history matching process, resulting in 26 parameters with initial ranges shown in Figure 7.

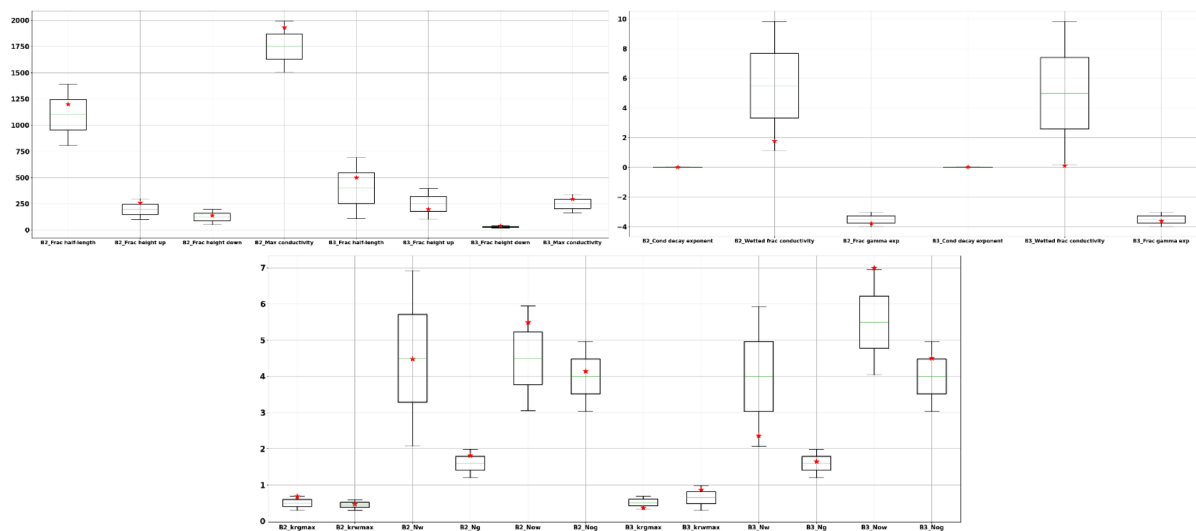


Figure 7—Fracture parameters (top two plots) and relative permeability parameters (bottom plot) for Benches B2 and B3 were selected as the uncertain variables in the history matching process, resulting in a total of 26 parameters. The parameters are displayed across separate plots to accommodate differences in scale. Red stars indicate the values used to compute the reference solution.

The production envelopes of the initial 30 trials are shown in Figure 8, illustrating a wide range of production outcomes. Initially, the MAPE of these trials was significantly higher than the modeling error between the surrogate and the simulator. After 20 iterations, the error relative to historical production was substantially reduced, as shown in Table 2 and Figure 8. The algorithm grouped the solutions into eight clusters, each with error levels comparable to the surrogate's modeling error relative to the simulator.

Table 2—Mean Absolute Percentage Error (MAPE) for each production phase in representative wells B2_0 and B3_1 over the one-year production history. For the phases that significantly contribute to total production—namely, all three phases in B2 wells and the water phase in B3 wells—each of the eight final clusters exhibits a MAPE substantially lower than the average MAPE observed across the initial 30 trials.

	B2_0			B3_1		
Case	Oil [%]	Water [%]	Gas [%]	Oil [%]	Water [%]	Gas [%]
Avg init trials	32	49	37	296,223	25	78
Cluster #1	9	4	4	64	11	42
Cluster #2	11	9	4	58	8	41
Cluster #3	12	6	13	81	10	41
Cluster #4	14	13	14	62	9	30
Cluster #5	12	10	7	64	9	40
Cluster #6	12	9	4	620	9	40
Cluster #7	11	10	5	62	9	42
Cluster #7	11	10	6	6	10	44

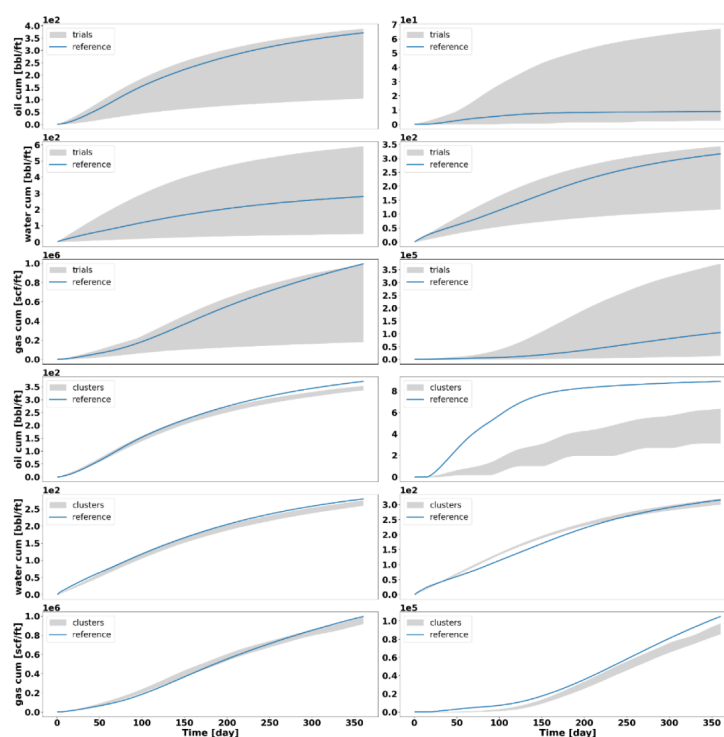


Figure 8—Range of production outcomes from the initial 30 history matching trials (top) and the final 8 clustered scenarios (bottom) for wells B2_0 (left) and B3_1 (right). Over the course of 20 main iterations, the trials progressively converge toward the historical production data. Several final scenarios fall within the modeling error of the surrogate relative to the reference simulator.

These eight clusters represent plausible development scenarios that can be used for long-term forecasting. Figure 9 compares the 40-year forecasts from these scenarios with the reference solution, showing relatively low uncertainty across the ensemble.

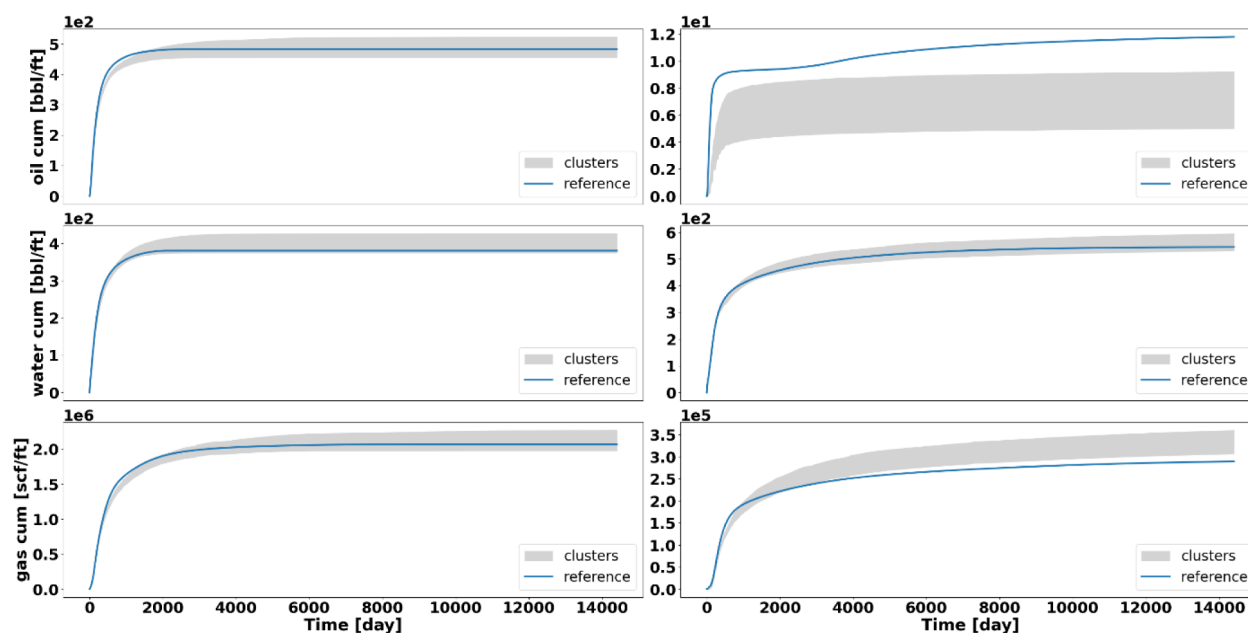


Figure 9—Range of production covered by the initial 8 scenarios for well B2_0 (left) and B3_1 (right) over a 40-year production period. All scenarios forecast production close to the reference solution for the 4 phases that influence overall production.

The entire process required only 2 hours and 35 minutes and involved 4,447 forecast evaluations using the surrogate model. In contrast, a single full-physics simulation of the same case takes approximately 30 minutes on 40 CPUs. While forecast evaluations could theoretically be run in parallel, synchronization is required between iterations. Given that the Gauss-Newton algorithm includes one sub-iteration per main iteration, a total of 40 synchronizations would be needed, resulting in a minimum of 20 hours to complete the process using a simulator—assuming access to 4,400 CPUs to run 110 simulations in parallel. If computational resources are limited to 1,000 CPUs, the process would take approximately 160 hours (nearly one week). Furthermore, any simulation delays due to convergence issues (e.g., from relative permeability non-linearities) would further extend the runtime, making the process impractical.

In contrast, the surrogate-based workflow enables completion of the history match in under half a day, allowing for multiple iterations within a single day if needed. This accelerates decision-making and enables the business to proceed to the next stage of the workflow with multiple validated scenarios—either continuing with the surrogate model or transferring the history-matched parameters to a full-physics simulator.

Conclusions

In this collaborative study, a lightning-fast reservoir surrogate model was developed to significantly reduce the cycle time associated with physics-based modeling workflows. The surrogate model captures subsurface uncertainty more comprehensively and benefits from key features of modern machine learning systems, including interpretability and confidence scoring, thereby providing users with richer insights to support decision-making.

The model's rapid forecasting capability was leveraged to identify multiple plausible development scenarios through an accelerated history matching process. When multiple solutions are sought, the number of forecast evaluations increases substantially making the process prohibitively time-consuming when using a traditional reservoir simulator. In contrast, the surrogate model consistently delivers forecasts with sufficient accuracy in a fraction of the time, enabling the generation of multiple scenarios within just a few hours.

These scenarios can then be seamlessly integrated into conventional reservoir development and depletion planning workflows, enhancing both efficiency and decision quality.

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